Cross-Domain Generalization and Evaluation Benchmarks for Scientific NLP Models

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ABSTRACT

State-of-the-art NLP models demonstrate impressive benchmark performance on scientific document understanding when evaluated within the same scientific discipline on which they were pre-trained. However, real-world research workflows routinely span interdisciplinary inquiries spanning materials science, bioinformatics, quantum computing, and economics. In this paper, we conduct an extensive multi-institutional study analyzing cross-domain generalization degradation across twelve specialized scientific domains. We introduce SciCross, a curated open-source benchmark consisting of 25,000 annotated document pairs with challenging cross-domain transfer tasks. Our findings demonstrate that standard transformers suffer an average performance drop of 34.6% under out-of-distribution domain shift, and we propose domain-invariant representation learning strategies to restore transfer robustness.

1. INTRODUCTION

Interdisciplinary scientific breakthroughs increasingly occur at the intersection of traditional academic boundaries?such as applying machine learning to drug discovery or computational physics to financial risk modeling. Consequently, automated research paper assistants must comprehend specialized terminology across heterogeneous domains without catastrophic degradation. While existing benchmarks like SciDocs and PubMed QA evaluate in-domain competence within computer science and biomedicine respectively, there is a distinct lack of standardized frameworks measuring cross-domain resilience.

2. BENCHMARK CORPUS DESIGN: SCICROSS

The SciCross benchmark spans twelve scientific disciplines: Computer Science, Biomedicine, High-Energy Physics, Materials Science, Chemical Engineering, Economics, Computational Neuroscience, Environmental Science, Mathematics, Astrophysics, Cognitive Science, and Quantum Computing.

Each domain contains:

- 1,000 full-text research papers with page-aware bounding boxes
- Curated entity pairs linking methodologies to experimental findings
- Cross-domain analogy queries requiring semantic alignment across divergent terminologies.

3. METHODOLOGY: DOMAIN-INVARIANT REPRESENTATION LEARNING

To counteract domain shift degradation, we propose InvariantSciNet, an architecture that decouples domain-specific stylistic vocabulary from universal scientific discourse logic (such as hypotheses, methodological formulations, and empirical controls).

We train the encoder using a multi-task objective:

L_total = L_task + lambda_1 * L_adversarial + lambda_2 * L_contrastive

where L_adversarial penalizes the representation if a domain classifier can successfully predict the source scientific field from latent embedding vectors, forcing the representation to learn domain-agnostic semantic geometries.

L_contrastive aligns analogous structural concepts (e.g., 'ablation study' in computer science and 'control group experiment' in clinical medicine) into close latent proximity.

4. EXPERIMENTAL RESULTS ACROSS DOMAINS

Table 2 details cross-domain zero-shot transfer accuracy.

Standard base models trained on Computer Science achieve 88.5% in-domain F1, but plunge to 52.1% when evaluated on Chemical Engineering and 49.8% on High-Energy Physics.

In contrast, InvariantSciNet maintains 76.4% on Chemical Engineering and 74.2% on Physics, cutting the cross-domain degradation margin by more than half.

Furthermore, retrieval systems utilizing domain-invariant embeddings achieve 22.3% higher Recall@10 when matching interdisciplinary literature.

5. FAILURE MODES AND REPORTED LIMITATIONS

Our error analysis reveals three persistent failure modes:

1. Highly specialized mathematical notation (e.g., Dirac bra-ket notation in Quantum Physics vs. tensor indices in Machine Learning) caused representation collapse in 14.2% of test cases.

2. Acronym collisions: ambiguous acronyms across disciplines (e.g., 'CNN' as Cellular Neural Network in neuroscience vs. Convolutional Neural Network in vision) frequently misdirect retrieval pipelines without explicit document context.

3. Heavy reliance on English-language publications; non-English scientific terminology remains entirely unaddressed in the current benchmark release.

6. CONCLUSION & FUTURE DIRECTIONS

We presented SciCross, demonstrating that cross-domain generalization is a pivotal frontier for AI research assistants. We urge the research community to adopt cross-domain stress-testing as standard protocol for evaluating scientific language models. Future releases will encompass multi-modal figures, tables, and multi-lingual preprints.

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